

UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH

Escola Superior d'Enginyeries Industrial,
Aeroespacial i Audiovisual de Terrassa

Universitat Politècnica de Catalunya

Escola Superior d'Enginyeries

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Pau Tusell Tresserras

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1 Introduction

The goal of this assignment is to analyse the interaction between a cylinder and air in a two dimensional space. In order to simplify the problem incompressible flow has been considered, thus forcing low air velocity.

The problem is composed of a duct, a solid cylinder located at the center and an inlet air flow.

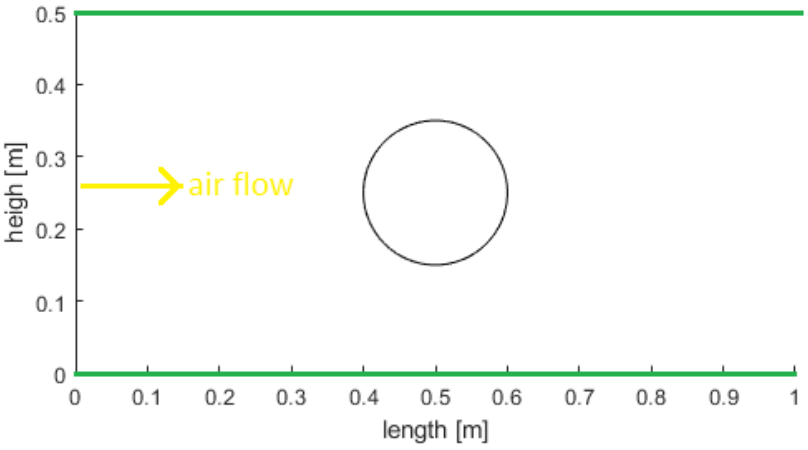


Figure 1: Representation of the problem situation

By means of a code, written in Matlab program, the most interesting coefficients for an aerospace engineer have been obtained as well as its representative plots considering three configurations:

- Channel. An empty duct has been computed in order to verify the values obtained.
- Static cylinder. This extension requires a larger code due to the addition of a centered cylinder.
- Rotating cylinder. The Psi of the center of the tunnel has been modified.

In all three situations the same inlet properties have been assumed:

- Pressure (P) = 101325 Pa
- Temperature (T) = 273 K
- Gas constant (R) = 287 J/kg K

2 Methodology

As stated, the results have been obtained numerically by means of a code. This implies the generation of a mesh, the evaluation of the equations at each control volume (or equivalently in this case at each interested point) and the post-process in order to obtain the plots.

It's very important at this point to note that two main directions have always been considered, being x and y:

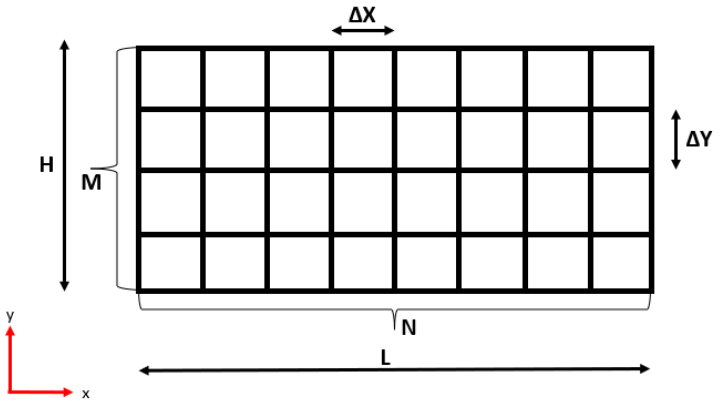


Figure 2: Representation of the mesh and the axis directions

2.1 Mesh

The discretization of the mesh is different in both sides, allowing an independent refinement if needed. This values are useful depending on the ratio between the height and the width of the tunnel. The characteristic parameters are:

- **L**: Length of the tunnel
- **H**: Height of the tunnel
- **N**: Number of divisions along the x direction
- **M**: Number of divisions along the y direction
- **ΔX**: Distance along the x direction of every division
- **ΔY**: Distance along the y direction of every division

It's remarkable that the dimension of each portion is proportional to the number of divisions:

$$\Delta X = \frac{L}{N} \qquad\qquad \Delta Y = \frac{H}{M}$$

The points of interest are located in the center of each control volume. Plus the boundary also has some extra interesting points. Those are located in the four vertex and at the center of each segment as can be seen.

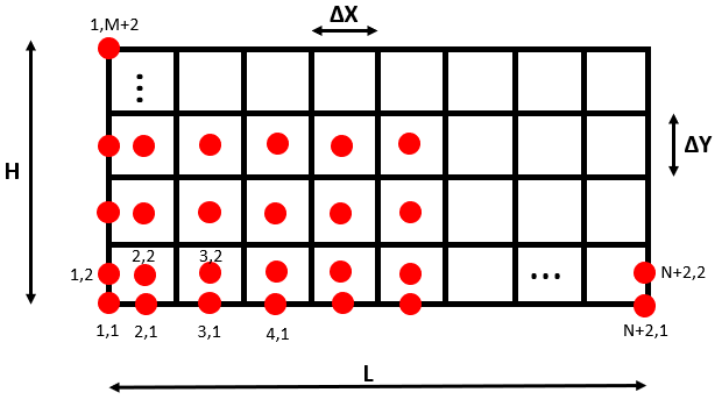


Figure 3: Representation of some of the nodes and their nomenclature

Here is important to explain the nomenclature that has been used. As the center of coordinates has been considered on the lower-left part and Matlab code operates with arrays being its initial position the number one, then each point has been named in order fulfilling with a (x,y) structure. A better understanding can be made by looking at the previous picture.

2.2 Boundary conditions

The four external parts take lot of importance during the calculations. For this reason each one has been represented in the following figure and has been explained below.

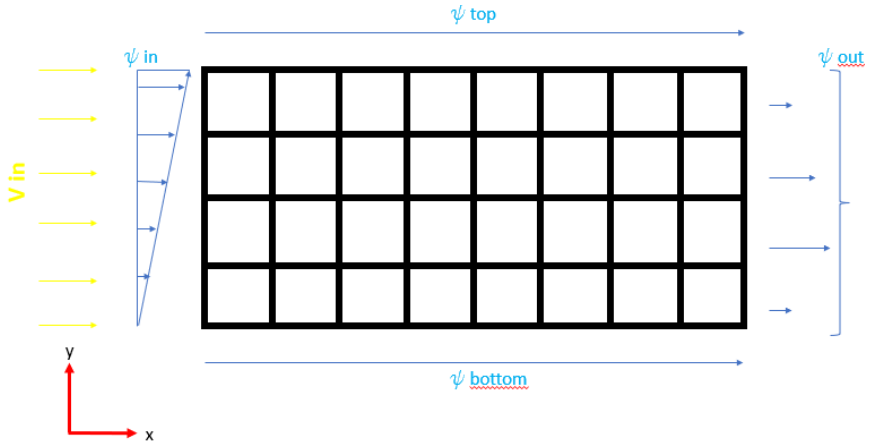


Figure 4: Stream functions at each boundary layer

- Vin: Inlet air flow. It's constant through all the section.
- ψ_{in}: Proportional to the velocity and the height of the desired node. It's equation is $\psi_{in} = Vin \cdot y$
- ψ_{top}: Same as the highest at the entry. Hence: $\psi_{top} = Vin \cdot H$
- ψ_{bottom}: Same as the lowest at the entry. Hence: $\psi_{bottom} = Vin \cdot 0 = 0$
- ψ_{out}: Same as the previous inlet node value. Hence: $\psi_{out} = \psi_{N+1}$

2.3 Internal nodes

As stated in the introduction, non-viscous effects have been considered through the entire domain. Moreover irrotational flow is present in all the region, making the flow irrotational ($\nabla x V = 0$).

The concept of stream functions has been used during all this problem as the only limitations are 2D and steady flow. Each stream function is characterised:

$$v_x = \frac{\rho_{ref}}{\rho}\frac{\partial \psi}{\partial y} \qquad v_y = -\frac{\rho_{ref}}{\rho}\frac{\partial \psi}{\partial x} \tag{1}$$

With the mentioned conditions then Stockes theorem can be written as:

$$\oint_C v \cdot dl = 0 \tag{2}$$

And circulation can now be computed.

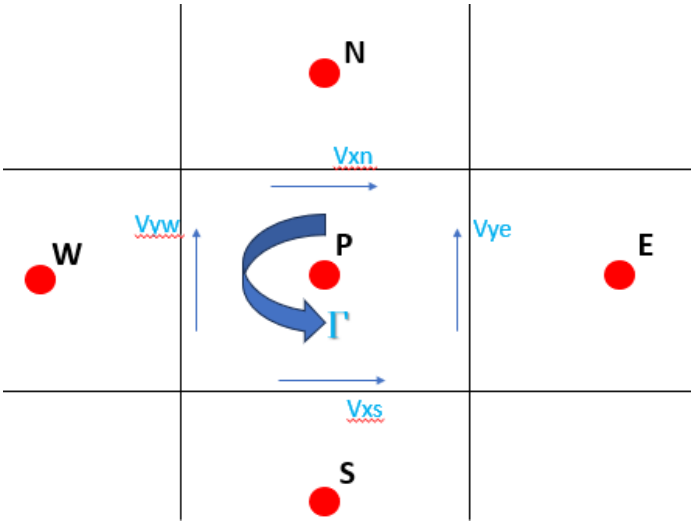


Figure 5: Circulation of a generic internal node

$$\Gamma \approx v_{ye} \cdot \Delta y - v_{xn} \cdot \Delta x - v_{yw} \cdot \Delta y + v_{xs} \cdot \Delta x = 0 \tag{3}$$

Taking into account the stream function definition (equation 1) and doing a second order approximation, then the circulation is as follows:

$$\Gamma \approx -\frac{\rho_{ref}}{\rho_e}\frac{\psi_E - \psi_P}{d_{PE}}\Delta y - \frac{\rho_{ref}}{\rho_n}\frac{\psi_N - \psi_P}{d_{PN}}\Delta x + \frac{\rho_{ref}}{\rho_w}\frac{\psi_P - \psi_W}{d_{PW}}\Delta y + \frac{\rho_{ref}}{\rho_s}\frac{\psi_P - \psi_S}{d_{PS}}\Delta x = 0 \tag{4}$$

This last equation can be rewritten by using coefficients so the code is more understanding.

$$a_P\psi_P = a_E\psi_W + a_W\psi_W + a_N\psi_N + a_S\psi_S + b_P. \tag{5}$$

For the internal nodes:

$$a_E = \frac{\rho_{ref}}{\rho_e}\frac{\Delta y}{d_{PE}}; a_W = \frac{\rho_{ref}}{\rho_w}\frac{\Delta y}{d_{PW}}; a_N = \frac{\rho_{ref}}{\rho_n}\frac{\Delta x}{d_{PN}}; a_S = \frac{\rho_{ref}}{\rho_s}\frac{\Delta x}{d_{PS}}; b_P = 0$$

2.4 Blocking-off method

When the cylinder is considered in the problem, the zone occupied by the figure must be treated different. Specifically the parts where there is a contact between a solid and a fluid region because the continuity of ψ is achieved but not its derivative.

At a generic face A (being the only possibilities E, W, N and S) the velocity obey the previous stated equation:

$$v_A = \frac{\rho_{ref}}{\rho_a}\frac{\psi_A - \psi_P}{d_{PA}} \tag{6}$$

Where the fraction of the densities is the calculated with the harmonic mean of the nodal values:

$$\frac{\rho_{ref}}{\rho_a} = \frac{d_{PA}}{\frac{d_{Pa}}{\rho_{ref}/\rho_P} + \frac{d_{Aa}}{\rho_{ref}/\rho_A}} \tag{7}$$

It's remarkable to note that this method has been used in all the nodes because it's a more generic expression. In other words, the code is written so that automatically the computer knows if the actual node is a fluid, a solid or a mix. In addition, a control volume is considered part of the solid if his central node is inside the cylinder. An example is shown below:

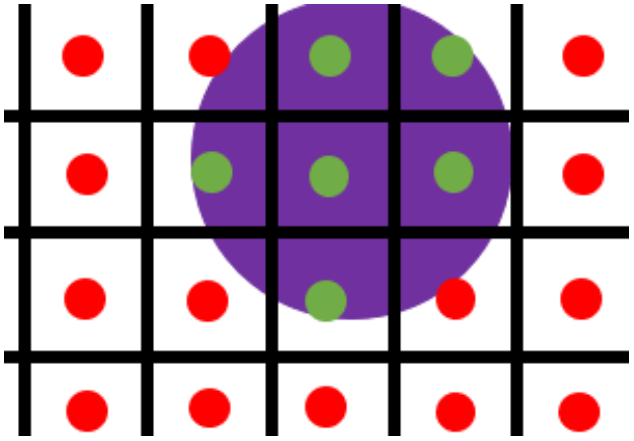


Figure 6: Representation of the blocking-off method

The control volumes that have their central node with green colour are the ones considered totally inside the cylinder. This is computed by comparing the cylinder radius with the distance from its center to the node.

2.5 Aerodynamic parameters

They are computed with the usual formulas. Notice in the code this values are only considered around the cylinder.

$$Cp(i,j) = 1 - \frac{V(i,j)^2}{V_{in}^2} \tag{8}$$

$$P = Cp \cdot 0.5 \cdot \rho_{in} \cdot V_{in}^2 + P_{in} \tag{9}$$

$$L = P \cdot \Delta x \tag{10}$$

$$D = P \cdot \Delta y \tag{11}$$

$$\Gamma = V_i \cdot \Delta i \tag{12}$$

Aerodynamic coefficients have been computed using the cylinder area as the section ($S = \pi \cdot r^2$):

$$Cl = \frac{L}{0.5 \cdot \rho_{in} \cdot V_{in}^2 \cdot S} \tag{13}$$

$$Cd = \frac{D}{0.5 \cdot \rho_{in} \cdot V_{in}^2 \cdot S} \tag{14}$$

3 Code

The following diagram represents the structure of the code, attached at the annex.

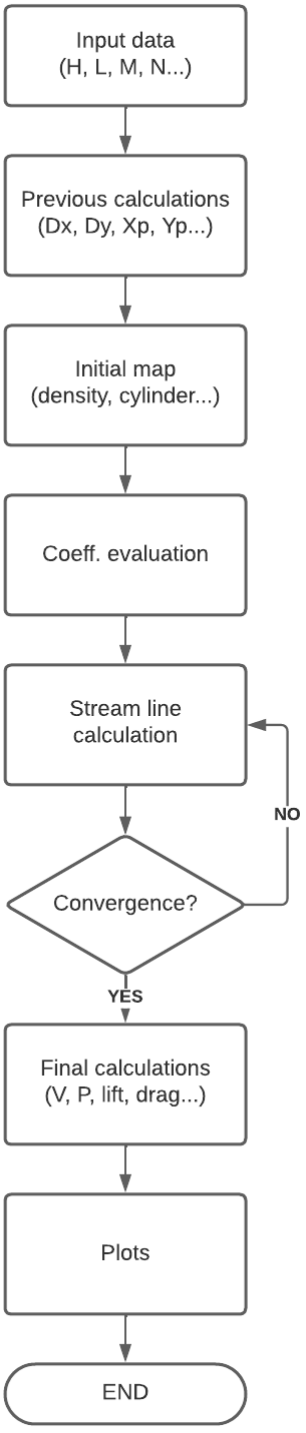


Figure 7: Code structure

A small description of each part is now done for a better understanding:

1. **Input data**. All required parameters are set. This involves the tunnel dimensions, the number

of mesh partitions, inlet conditions, cylinder radius...

2. **Previous calculations**. Mesh parameters are calculated such as the control volume dimensions, and the position of the nodes.
3. **Initial map**. Matrix dimensions of the density, a coefficients and location of the cylinder.
4. **Coeff evaluation**. Calculation of all the coefficients a and density ratios (with the previously explained harmonic mean).
5. **Stream line calculation**. ψ_P is obtained at each point with equation 5. As it's an iterative process, every time the resulting ψ_P must be compared with the previous one and the process ends when the maximum difference is smaller than the stated delta value.
6. **Final calculations**. With the stream lines all the interesting values can be obtained: lift, drag, pressure and circulation.
7. **Plots**. The representation of the results allow a better understanding of the situation. Hence the tunnel along with each result and the cylinder has been plotted at the same time. See next section for further information.

4 Verification

The code itself can't be checked. However some of the results can be compared with real solutions. This is the case of a channel, so the tunnel without the cylinder has been computed:

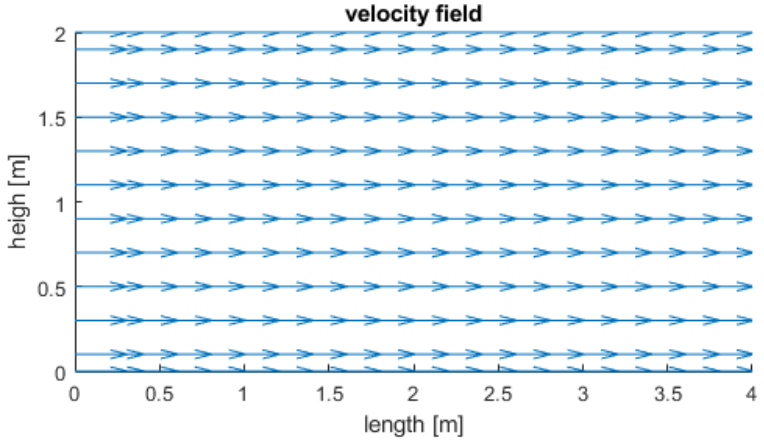


Figure 8: Air velocity representation through an empty channel

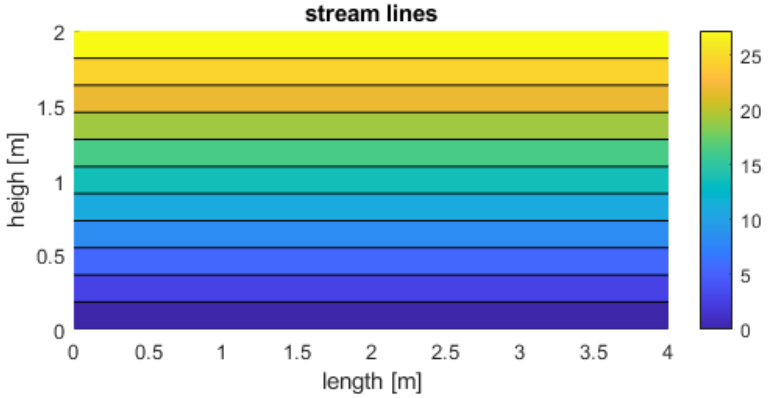


Figure 9: Stream lines through an empty channel

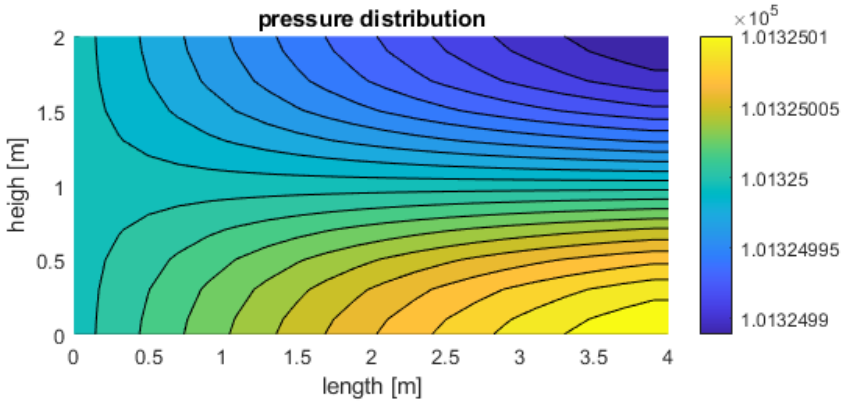


Figure 10: Pressure (Pa) contour isolines through an empty channel

The velocity field is constant in speed value (same as the inlet one) and direction. This result is the one that was expected because it's incompressible air going through an empty channel.

Stream lines are also horizontal and their value varies according to the inlet boundary condition $(v \cdot h)$, so one concludes that this result was also expected.

For the pressure distribution one could think that it has been computed wrong. However taking a deep look the variation between the values is found at the thousandth. This minor deviation comes from the code when computing the results so this effect is considered negligible.

In conclusion for this part, all three plots have the desired results and shapes. Hence it can be assumed the code works properly.

5 Results

First the most interesting parameters are resumed for each case. Note they are all the same (except the mesh divisions) for a better comparison of the results.

	N	M	L (m)	H (m)	radius (m)	delta
channel	20	10	4	2	0.5	1e-5
static cyl.	20	10	4	2	0.5	1e-5
rotating cyl.	20	10	4	2	0.5	1e-5
rotating cyl. with refin.	60	40	4	2	0.5	1e-5

Table 1: Table comparing the problems' main characteristic values

Inlet conditions have always been Vin = 15 m/s, Tin = 293 K, Pin = 101325 Pa and the density of the cylinder almost 0 (1e-10).

5.1 Channel with a static cylinder

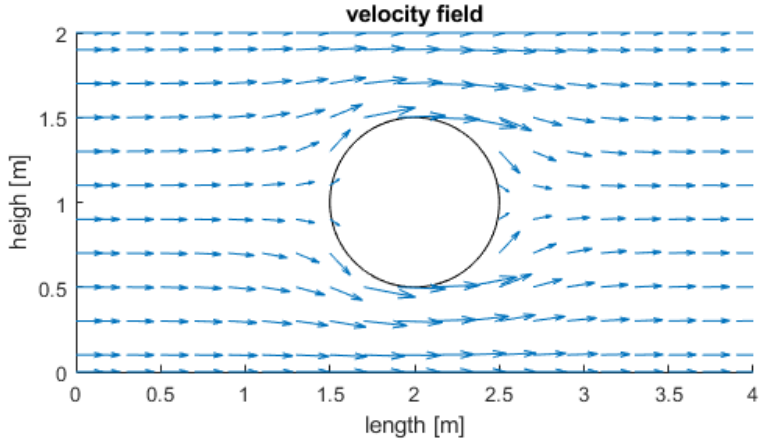


Figure 11: Air velocity representation through a channel with a centered static cylinder

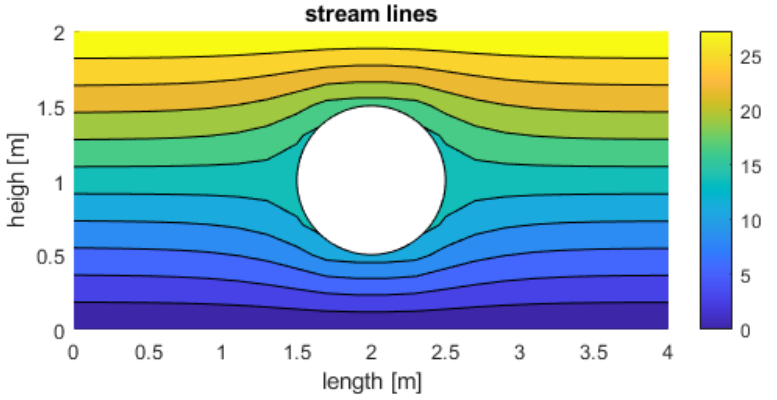


Figure 12: Stream lines through a channel with a centered static cylinder

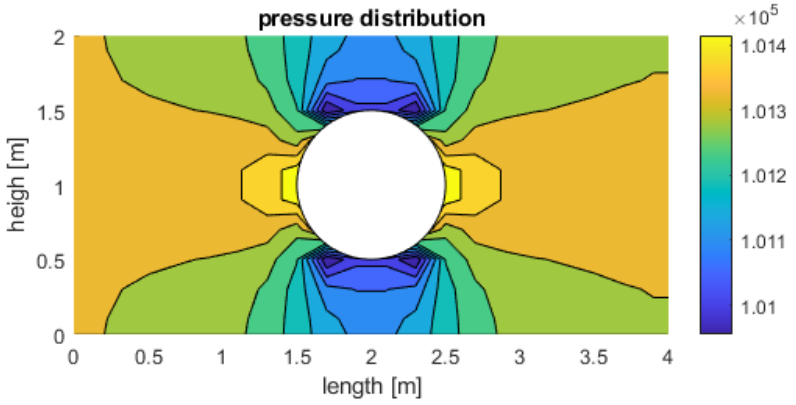


Figure 13: Pressure (Pa) contour isolines through a channel with a centered static cylinder

With the velocity and the stream lines plots one can guarantee that air surrounds the cylinder symmetrically, as expected. In addition at the top and bottom external part of the cylinder velocity increase while pressure decrease. This is also a known and expected behaviour.

5.2 Channel with a rotatory cylinder

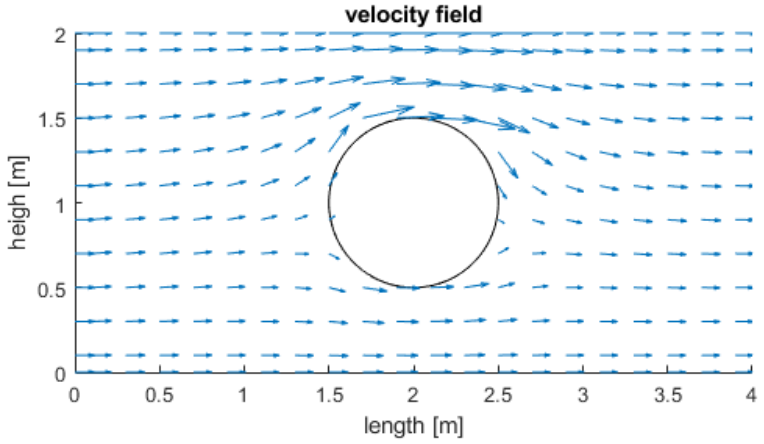


Figure 14: Air velocity representation through a channel with a centered rotating cylinder

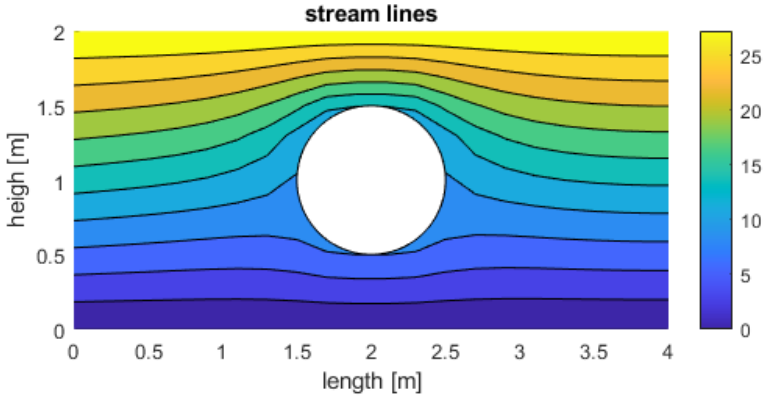


Figure 15: Stream lines through a channel with a centered rotating cylinder

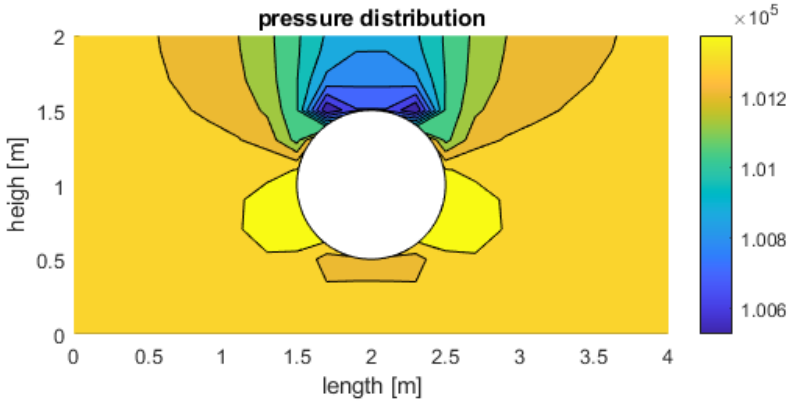


Figure 16: Pressure (Pa) contour isolines through a channel with a centered rotating cylinder

Again the behaviour of the flow is the expected one. Air speed is increased on the top part of the rotating cylinder, creating a pressure drop. Hence lift is generated. Notice the stream lines plot also indicate the same mentioned pattern in the flow.

5.3 Refinements of the channel with a rotatory cylinder

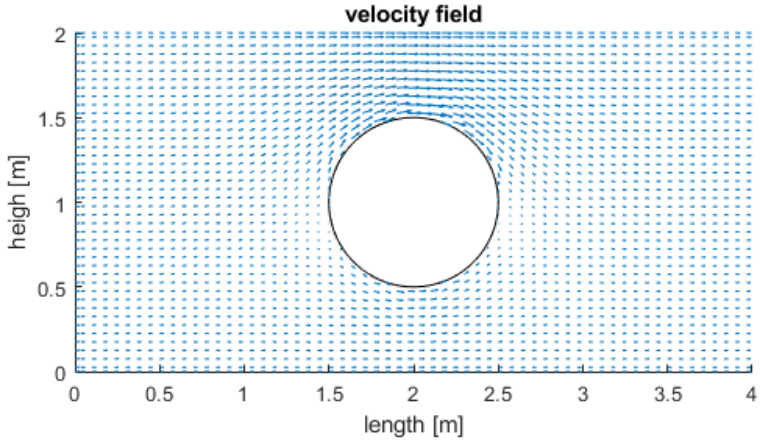


Figure 17: Air velocity representation through a channel with a centered rotating cylinder and a refined mesh

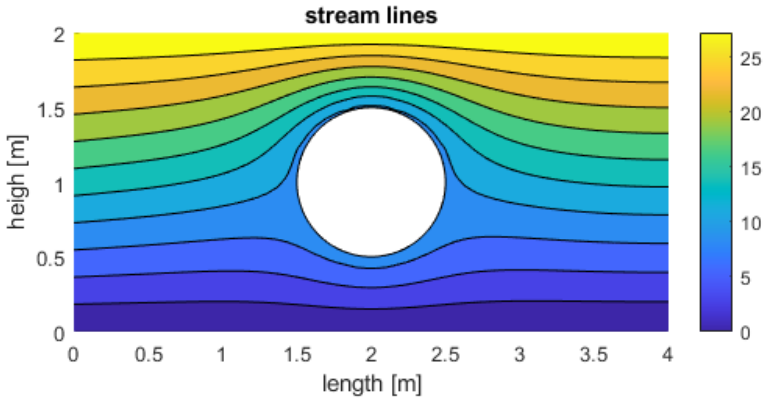


Figure 18: Stream lines through a channel with a centered rotating cylinder and a refined mesh

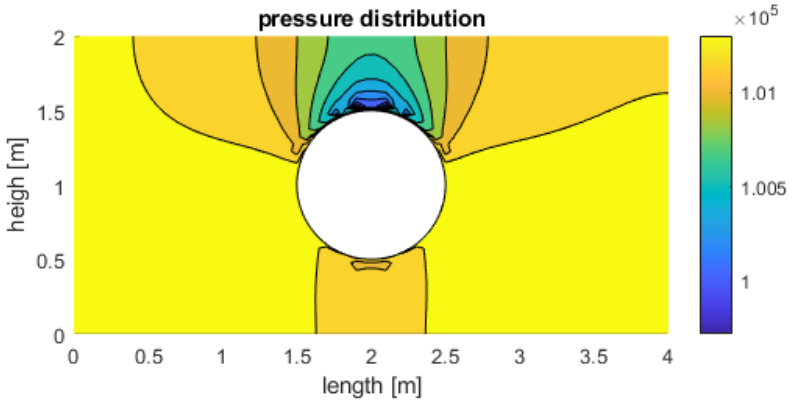


Figure 19: Pressure (Pa) contour isolines through a channel with a centered rotating cylinder and a refined mesh

The pressure plot shows the same distribution and values compared to the case of a less refined mesh.

Stream lines are also the same ones, being the only difference the smoothness due to the mentioned better resolution of the mesh.

Last but not least from this section there is the velocity field which represents the same scenario: higher velocity in the upper part.

5.4 Aerodynamic results

Cl	Cd	Γ
4.9246	-0.0144	30.5744

Table 2: Main aerodynamic results

Lift coefficient seems to be pretty good regarding the 15 m/s inlet velocity and drag is two orders of magnitude below, as it should be. The minus sign indicates the force direction points to the left (negative x direction). This results are for the less refined rotatory case.

6 Conclusion

Given an inlet speed and flow characteristics this code has proven the behaviour of the air inside an empty channel as well as a channel with a cylinder. In all cases the results were the expected ones: parallel flow in the first case, symmetrical distributions for the static cylinder and pressure as well as velocity variations in the upper part for the rotatory case. Plus the refinement gave an accurate result, but the difference was so slight that is not worthy.

Finally, for the rotatory case, the main aerodynamic parameters proved lift is generated and drag is also present with a smaller magnitude.

7 Annex

```
%
clear all
clc

%----input data----
L = 4; %m
H = 2; %m
Tin = 273+20; %K
Pin = 101325; %Pa
R = 287;
Vin = 15; %m/s

N=20;
M=10;
delta=1e-5;
PSIstart = 1;
acabar = 0;

radius = 0.5; %m (cylinder radius)

cylinder_exists = 1;   % if 1 then it exists. if 0 then it don't exist
rotating_cylinder = 1; % if 1 then it rotate. if 0 then it don't rotate

%----previous calculations----
RHOin = Pin/(Tin*R); %kg/m3

deltaX = L/N;
deltaY = H/M;

Xcv = zeros(N+1,1);
Xp = zeros(N+2,1);
Ycv = zeros(M+1,1);
Yp = zeros(M+2,1);

for i=1:1:N+1
    Xcv(i,1) = deltaX*(i-1);
end

for j=1:1:M+1
    Ycv(j,1) = deltaY*(j-1);
end

Xp(1,1) = 0;
Xp(N+2,1) = L;
for i=2:1:N+1
    Xp(i,1) = deltaX/2 + deltaX*(i-2);
end

Yp(1,1) = 0;
Yp(M+2,1) = H;
for j=2:1:M+1
    Yp(j,1) = deltaY/2 + deltaY*(j-2);
end

%----initial map----
RHO = zeros(N+2,M+2);
RHO(:,:) = RHOin;
PSI = zeros(N+2,M+2);
PSIestimate = zeros(N+2,M+2);
PSIestimate(:,:) = PSIstart;
```

```
62
63 aE = zeros(N+2,M+2);
64 aW = zeros(N+2,M+2);
65 aS = zeros(N+2,M+2);
66 aN = zeros(N+2,M+2);
67 bp = zeros(N+2,M+2);
68
69 %----calculation of the cylinder-----
70 solid_region = zeros(N+2,M+2);
71 cylinderCENTRE=[L/2 H/2];
72 if cylinder_exists==1
73     for i=1:1:N+2
74         for j=1:1:M+2
75             distTOcylinderCENTRE = sqrt( (Xp(i,1)-cylinderCENTRE(1,1))^2 + (Yp(j
    ,1)-cylinderCENTRE(1,2))^2 );
76             if distTOcylinderCENTRE <= radius
77                 solid_region(i,j)=1;
78                 RHO(i,j)=1e-10;
79             end
80         end
81     end
82 end
83
84 %----evaluation of coefficients----
85 %bottom
86 aE(:,1)=0; aW(:,1)=0; aN(:,1)=0; aS(:,1)=0; bp(:,1)=0; aP(:,1)=1;
87 PSIestimate(:,1)=0;
88 %top
89 aE(:,M+2)=0; aW(:,M+2)=0; aN(:,M+2)=0; aS(:,M+2)=0; bp(:,M+2)=Vin*H; aP(:,M+2)=1;
90 PSIestimate(:,M+2)=bp(:,M+2);
91 %inlet
92 aE(1,:)=0; aW(1,:)=0; aN(1,:)=0; aS(1,:)=0; aP(1,:)=1;
93 for j=1:1:M+2
94     bp(1,j) = Vin*Yp(j,1);
95     PSIestimate(1,j)=bp(1,j);
96 end
97 %internal
98 for i=2:1:N+1
99     for j=2:1:M+1
100         dPE(i,j) = deltaX;
101         dPe = deltaX/2;
102         dEe = deltaX/2;
103
104         dPW(i,j) = deltaX;
105         dPw = deltaX/2;
106         dWw = deltaX/2;
107
108         dPN(i,j) = deltaY;
109         dPn = deltaY/2;
110         dNn = deltaY/2;
111
112         dPS(i,j) = deltaY;
113         dPs = deltaY/2;
114         dSs = deltaY/2;
115
116         if i==2 && j==2
117             dPW(i,j) = deltaX/2;
118             dPw = deltaX/2;
119             dWw = 0;
120             dPS(i,j) = deltaY/2;
121             dPs = deltaY/2;
122             dSs = 0;
123         elseif i==N+1 && j==2
```

```
            dPS(i,j) = deltaY/2;
            dPs = deltaY/2;
            dSs = 0;
            dPE(i,j) = deltaX/2;
            dPe = deltaX/2;
            dEe = 0;
        elseif i==2 && j==M+1
            dPW(i,j) = deltaX/2;
            dPw = deltaX/2;
            dWw = 0;
            dPN(i,j) = deltaY/2;
            dPn = deltaY/2;
            dNn = 0;
        elseif i==N+1 && j==M+1
            dPN(i,j) = deltaY/2;
            dPn = deltaY/2;
            dNn = 0;
            dPE(i,j) = deltaX/2;
            dPe = deltaX/2;
            dEe = 0;
        elseif i==2
            dPW(i,j) = deltaX/2;
            dPw = deltaX/2;
            dWw = 0;
        elseif i==N+1
            dPE(i,j) = deltaX/2;
            dPe = deltaX/2;
            dEe = 0;
        elseif j==2
            dPS(i,j) = deltaY/2;
            dPs = deltaY/2;
            dSs = 0;
        elseif j==M+1
            dPN(i,j) = deltaY/2;
            dPn = deltaY/2;
            dNn = 0;
        end

        RHOref_RHOe(i,j) = dPE(i,j)/((dPe/(RHOin/RHO(i,j)))+(dEe/(RHOin/RHO(i+1,j)
    )));
        RHOref_RHOw(i,j) = dPW(i,j)/((dPw/(RHOin/RHO(i,j)))+(dWw/(RHOin/RHO(i-1,j)
    )));
        RHOref_RHOn(i,j) = dPN(i,j)/((dPn/(RHOin/RHO(i,j)))+(dNn/(RHOin/RHO(i,j+1)
    )));
        RHOref_RHOs(i,j) = dPS(i,j)/((dPs/(RHOin/RHO(i,j)))+(dSs/(RHOin/RHO(i,j-1)
    )));


        aE(i,j) = RHOref_RHOe(i,j)*(deltaY/dPE(i,j));
        aW(i,j) = RHOref_RHOw(i,j)*(deltaY/dPW(i,j));
        aN(i,j) = RHOref_RHOn(i,j)*(deltaX/dPN(i,j));
        aS(i,j) = RHOref_RHOs(i,j)*(deltaX/dPS(i,j));
        aP(i,j) = aE(i,j)+aW(i,j)+aN(i,j)+aS(i,j);
        bp(i,j) = 0;
    end
end
%outlet
aE(N+2,:)=0; aW(N+2,:)=1; aN(N+2,:)=0; aS(N+2,:)=0; bp(N+2,:)=0; aP(N+2,:)=1;
PSIestimate(N+2,:)=PSIestimate(N+1,:);
PSI=PSIestimate;

while acabar == 0
    for i=2:1:N+1
```

```
        for j=2:1:M+1
            if  cylinder_exists==1 && rotating_cylinder==1 && solid_region(i,j)==1
                PSI(i,j) = (Vin*(H/2))-5;
            elseif cylinder_exists==1 && rotating_cylinder==0 && solid_region(i,j)
    ==1
                PSI(i,j) = Vin*(H/2);
            else
                PSI(i,j)=(aE(i,j)*PSIestimate(i+1,j) + aW(i,j)*PSIestimate(i-1,j)
    + aN(i,j)*PSIestimate(i,j+1) + aS(i,j)*PSIestimate(i,j-1) + bp(i,j))/aP(i,j);
            end
        end
    end
    PSI(N+2,:)=PSI(N+1,:);
    %PSI(:,8)=Vin*H*0.5;

    if max( abs(PSIestimate-PSI) ) < delta
        acabar = 1;
    else
        PSIestimate=PSI;
    end
end


Vxp = zeros(N+2,M+2);
Vyp = zeros(N+2,M+2);
Vp = zeros(N+2,M+2);


for i=2:1:N+1
    for j=2:1:M+1

        Vxn(i,j) = RHOref_RHOn(i,j)*((PSI(i,j+1)-PSI(i,j))/(dPN(i,j)));
        Vxs(i,j) = -RHOref_RHOs(i,j)*((PSI(i,j-1)-PSI(i,j))/(dPS(i,j)));
        Vye(i,j) = -RHOref_RHOe(i,j)*((PSI(i+1,j)-PSI(i,j))/(dPE(i,j)));
        Vyw(i,j) = RHOref_RHOw(i,j)*((PSI(i-1,j)-PSI(i,j))/(dPW(i,j)));

        Vxp(i,j) = (Vxn(i,j)+Vxs(i,j))/2;
        Vyp(i,j) = (Vye(i,j)+Vyw(i,j))/2;

        Vp(i,j) = sqrt(Vxp(i,j)^2 + Vyp(i,j)^2);

    end
end



for i=1:1:N+2
    for j=1:1:M+2

        if i==1
            Vp(i,j) = Vin;
            Vxp(i,j) = Vin;
            Vyp(i,j) = 0;

        elseif i==N+2
            Vp(i,j) = Vp(i-1,j);
            Vxp(i,j) = Vxp(i-1,j);
            Vyp(i,j) = Vyp(i-1,j);

        elseif j==1
            Vp(i,j) = Vxs(i,j+1);
            Vxp(i,j) = Vxs(i,j+1);
            Vyp(i,j) = 0;
```

```
244
245         elseif j==M+2
246             Vp(i,j) = Vxn(i,j-1);
247             Vxp(i,j) = Vxn(i,j-1);
248             Vyp(i,j) = 0;
249
250         elseif solid_region(i,j)==1
251             Vp(i,j) = 0;
252             Vxp(i,j) = 0;
253             Vyp(i,j) = 0;
254         end
255     end
256 end
257
258
259
260 figure(1)
261 hold on
262 if cylinder_exists==1
263 theta = 0:0.01:2*pi;
264 fill(radius*cos(theta)+cylinderCENTRE(1,1), radius*sin(theta)+cylinderCENTRE(1,2),
        'w');
265 end
266 axis equal
267 xlim([0 L]);
268 ylim([0 H]);
269 Xpos=zeros(N+2,M+2);
270 for ii=1:1:M+2
271     Xpos(:,ii)=Xp(:,1);
272 end
273 Ypos=zeros(N+2,M+2);
274 for jj=1:1:M+2
275     Ypos(:,jj)=Yp(jj,1);
276 end
277 quiver(Xpos,Ypos,Vxp,Vyp)
278 title('velocity field');
279 xlabel('length [m]');
280 ylabel('heigh [m]');
281 hold off
282
283
284 Cp = zeros(N+2,M+2);
285 P = zeros(N+2,M+2);
286
287 Cp(:,:)=1-((Vp(:,:).^2)/(Vin^2));
288 P(:,:) = Cp(:,:)*0.5*RHOin*Vin^2+Pin;
289 L=0;
290 D=0;
291 Circulation = 0;
292
293 for i=2:1:N+1
294     for j=2:1:M+1
295
296         if solid_region(i+1,j)==1 && solid_region(i-1,j)==0 && solid_region(i,j+1)
        ==0 && solid_region(i,j-1)==0
297             D=D+P(i,j)*deltaY;
298             Circulation = Circulation + Vye(i,j)*deltaY;
299         elseif solid_region(i+1,j)==0 && solid_region(i-1,j)==1 && solid_region(i,
        j+1)==0 && solid_region(i,j-1)==0
300             D=D-P(i,j)*deltaY;
301             Circulation = Circulation - Vyw(i,j)*deltaY;
302         elseif solid_region(i+1,j)==0 && solid_region(i-1,j)==0 && solid_region(i,
        j+1)==1 && solid_region(i,j-1)==0
```

```
303             L=L+P(i,j)*deltaX;
304             Circulation = Circulation - Vxn(i,j)*deltaX;
305         elseif solid_region(i+1,j)==0 && solid_region(i-1,j)==0 && solid_region(i,
    j+1)==0 && solid_region(i,j-1)==1
306             L=L-P(i,j)*deltaX;
307             Circulation = Circulation + Vxs(i,j)*deltaX;
308         end
309
310     end
311 end
312
313 Cl=L/(0.5*RHOin*Vin^2*pi*radius^2);
314 Cd=D/(0.5*RHOin*Vin^2*pi*radius^2);
315
316
317
318 figure(2)
319 hold on
320 contourf(Xpos,Ypos,PSI,10);
321 colorbar;
322 if cylinder_exists==1
323 theta = 0:0.01:2*pi;
324 fill(radius*cos(theta)+cylinderCENTRE(1,1), radius*sin(theta)+cylinderCENTRE(1,2),
    'w');
325 end
326 axis equal
327 title('stream lines');
328 xlabel('length [m]');
329 ylabel('heigh [m]');
330 hold off
331
332
333
334 figure(3)
335 hold on
336 contourf(Xpos,Ypos,P,10);
337 colorbar;
338 if cylinder_exists==1
339 theta = 0:0.01:2*pi;
340 fill(radius*cos(theta)+cylinderCENTRE(1,1), radius*sin(theta)+cylinderCENTRE(1,2),
    'w');
341 end
342 axis equal
343 title('pressure distribution');
344 xlabel('length [m]');
345 ylabel('heigh [m]');
346 hold off
```